CADFEM-Bench v1.5: published-source CAD + FEM pilot

2026-10-14. **46 active tasks** with **95 graded outputs** in 12 engineering families; 51 tasks held with reasons; 190 locally computed v1.0 targets excluded. Every graded target is a value published by its source (saved solver results, publisher tables and evaluators, result legends, NAFEMS targets); none was computed for this release. This is a **source-backed pilot**, not a certified benchmark: 15 tasks are independently reproduced, most tolerances are provisional, and 15 of 46 tasks come from three parametric families (SimJEB, AgentFEM, GeoPINONet). Frictional contact, plasticity, forming, crash, fatigue and coupled assembly mechanics are not covered by the active tasks.

Changes in v1.5. Two physics classes that were missing from the active tasks are added, each graded only after an independent solver run settles within a 10 % band around the published value. (1) **Plasticity**: 1 fcVM elastic-perfectly plastic collapse (limit) load factors, checked by an independent FEniCSx limit analysis; the tensile bar (closed form) is calibration. (2) **Fracture**: 4 Code_Aster SSLP311 tasks, energy release rates at both tips of a crack on an inclined bimaterial interface (published boundary-element reference), checked by an independent J-integral study. All v1.4 tasks and targets are unchanged. Held after review: A-FCVM-PL-transition; the Hertz-contact and indentation cases, whose analytical references do not fit the posed geometry; frictional contact remains absent.

Release v1.4: balanced primary suite. Models receive only the primary suite: at most five tasks per engineering family, so the bracket and holed-plate families contribute 15 of 46 tasks instead of 84 of 126. The rest of the v1.3 pool is kept in evaluator-only suites that are scored separately, never added to the primary score. All targets, tolerances and task texts are the v1.3 ones.

suite	tasks	graded outputs	role	release check (this build)
primary (candidate/)	46	95	the scored comparison suite	exact answers 46/46 numerical, 0 passes; x1.3 answers 0; 6 geometry gates probed
calibration	7	7	NAFEMS controls; report separately	exact answers 7/7 numerical, 0 passes; x1.3 answers 0; 3 geometry gates probed
reserved-public	78	215	public-source tasks kept out of model inputs (not a private hold-out)	exact answers 78/78 numerical, 0 passes; x1.3 answers 0
related-public	1	1	shares CAD with a primary task	exact answers 1/1 numerical, 0 passes; x1.3 answers 0
full pool (task-pool/)	132		v1.3 tasks and references, unchanged	

Execution and field checks. *evaluator/sandbox_runner.py* freezes a submission and its inputs and replays main.py offline. A denial probe re-run against this folder (macos-seatbelt) was blocked from reading this package, the released v1.3 references and workspace source data, writing outside its job, modifying inputs and opening network connections; the pinned CadQuery/Gmsh/NumPy/SciPy runtime still imported. A trusted software solver replayed the cycloidal coupler in the sandbox: maximum displacement 0.00770554 mm on 12328 elements (published 0.00761747 mm, tolerance 6 %). *verify_static_fields.py* re-assembles the source-defined operator: relative residual 9.1e-13 for that field, and 1.0e-03 (rejected) for the same field scaled by 1.001. These are infrastructure checks and one solver baseline, not model or expert runs; boundary-face scope, geometry and convergence still need the trusted review. The common run protocol (budgets, attempts, provenance, separate numerical/execution/geometry/workflow metrics) is in *evaluator/evaluation-protocol.json*.

Tracks. A: the model receives the CAD and the source set-up and must compute the requested results (40 tasks). B: the model receives a dimensioned drawing or written specification, builds the CAD and analyses it (6 tasks; the built part is gated against an evaluator-only reference solid). **Tiers.** source_backed_pilot (numerical published reference), figure_reference (legend value, set-up partly from the source's BC figure), control (NAFEMS).

engineering family	tracks	tiers	tasks	graded outputs	distinct CAD	physics	reference class
agentfem-bracket	A	source_backed_pilot	5	10	5	linear static	published_table
cycloidal-drive-components	A	source_backed_pilot	5	5	5	linear static	published_raw_field
femcad-hard	B	source_backed_pilot	5	7	0	Electromagnetics, Heat Transfer, Solid Mechanics	closed_form, published_table
geopinonet-holed-plate	A	source_backed_pilot	5	10	5	linear static	published_raw_field
morphing-wing-plates	A	source_backed_pilot	5	15	5	modal	published_table
rigging-plate	A/B	figure_reference	5	5	1	linear static	published_legend
simjeb-bracket	A	source_backed_pilot	5	20	5	linear static, multi-load-case	published_table

engineering family	tracks	tiers	tasks	graded outputs	distinct CAD	physics	reference class
fcvm-buckling	A	source_backed_pilot	4	4	4	eigenvalue, linear buckling	published_table
interface-crack-fracture	A	source_backed_pilot	4	8	4	bimaterial interface crack, linear elastic fracture mechanics, plane stress	published_table
cadworld-freecad-fem	A	source_backed_pilot	1	4	1	linear static	publisher_evaluator
feenox-valve-transient	A	source_backed_pilot	1	6	1	multi-material, nonlinear material, transient thermal	published_table
fcvm-plastic-limit	A	source_backed_pilot	1	1	1	elastoplastic, limit analysis, von Mises perfect plasticity	published_table
total			46	95	37		

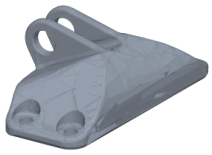
Changes in v1.3 (2026-10-14). (1) The 8 drawing/specification-to-CAD tasks held in v1.1 for lack of a geometry check are restored. Each has an evaluator-only reference solid built from the task's own specification (NAFEMS LE10: the source CAD) and accepted only if its volume matches a closed form or Pappus's theorem (worst disagreement 3e-08). The submitted part must match it within 2 % symmetric volume difference after alignment. (2) New family: fcVM elastic buckling. These are 8 FreeCAD steel models with the source's published outputs. The load set rebuilt from each model matches the published load sum, and an independent FEniCSx/SLEPc eigenvalue study admits the published first factor only if it settles within the cycloidal rule: 4 admitted, 4 held. Plastic-collapse and post-buckling factors from the same source are held. (3) Registry counts are recomputed after every hold.

Changes in v1.2 (2026-10-14). The published cycloidal displacements are coarse-mesh values (the coupler converges about 5 % above its reference, outside the old 5 % tolerance). An independent FEniCSx convergence study now sets each part's tolerance to the observed published-vs-converged difference plus twice the last refinement change (at least 5 %); 5 parts are admitted this way and 4 held. Published values are unchanged. AgentFEM fresh_test_000 is recorded as reproduced (follow-up study).

Changes from v1.0 (audit of 2026-10-13). (1) Scoring separates *numerical_passed* from *workflow_verified*; a pass needs both, and workflow verification needs eight artifacts (results, script, STEP, mesh, fields, solver log, refinement study, environment) plus a trusted review bound to task, run and artifact hashes. (2) The 190 targets computed by the local SimJEB grader are removed; SimJEB analysis keeps SimJEB's published displacement maxima and the 40 design tasks are held. (3) The two SimJEB STEP files that fail OpenCASCADE validity are held. (4) The rigging drawing task has a volume-overlap gate against the original CAD after alignment over the 24 axis rotations; the other 8 reconstruction tasks are held. (5) CADWorld's 'Tresca' column is (s1 - s3)/2 and is now requested as maximum shear stress. (6) Malformed submissions are rejected per task without stopping the run. (7) Tiers are renamed to what they show, and the original source files for every reference ship in an evaluator-only evidence bundle with hashes.

Verification at build time. 74 input files and 544 evidence files hash-checked; all 37 distinct CAD files valid in OpenCASCADE; no reference number in any prompt or text input; no verifier target. Scorer probes: exact answers alone give 46 numerical passes and 0 passes; answers scaled by 1.3 give 0 numerical passes; four malformed submissions are rejected without stopping the run; a reviewed probe passes and is rejected once an artifact changes; the 6 geometry gates accept their reference solid rotated and moved (symmetric difference at most 6.6e-14) and reject it scaled by 3 % (at least 0.246; limit 0.02; 5 are rejected earlier by the solid-count or analytic-radius check); submitting the independent finest-mesh values passes all 14 convergence-study tasks. Earlier set-up checks still hold: cycloidal fixed node sets lie on the stated faces and per-face loads sum to the deck resultants; GeoPINONet section forces are -1000 N; AgentFEM CAD scaled to mm matches the FEM nodes; the valve's Gmsh surface tags equal the OpenCASCADE face indices.

GE jet-engine bracket: four-case analysis of real competition designs



Ti-6Al-4V bracket, four GE load cases at the clevis pin (RBE3), rigid fully fixed bolts. Output per case: maximum displacement magnitude, graded against SimJEB's published OptiStruct maxima (10 %). Peak stress and the pass verdict are no longer graded (SimJEB does not publish them). Held: analyses 15 and 120 (invalid STEP) and all 40 design/redesign/lighten tasks (graded by the local FEM grader).

Tasks: 5. **Licence:** Dataverse CC0 / README ODC-By; GE bracket CAD from GrabCAD (non-commercial). Local research use. **Source:** SimJEB: simulated jet engine bracket dataset (Whalen, Beyene, Mueller, CGF 2021) / GE bracket challenge.

Example A-SIMJEB-analysis-0 (hidden references; 1 evidence files):

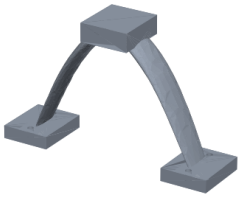
output	unit	reference	tolerance	class	evidence / provenance
vertical_max_displacement_mm	mm	0.321826	rel 0.1	published_table	SimJEB metadata CSV id 0 column max_ver_magdisp (OptiStruct)
horizontal_max_displacement_mm	mm	0.239873	rel 0.1	published_table	SimJEB metadata CSV id 0 column max_hor_magdisp (OptiStruct)
diagonal_max_displacement_mm	mm	0.165667	rel 0.1	published_table	SimJEB metadata CSV id 0 column max_dia_magdisp (OptiStruct)
torsion_max_displacement_mm	mm	0.174951	rel 0.1	published_table	SimJEB metadata CSV id 0 column max_tor_magdisp (OptiStruct)

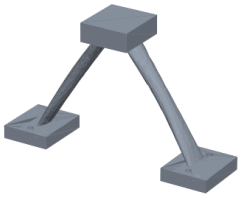
Example prompt (A-SIMJEB-analysis-0), first lines:

```
Design task: the GE jet engine bracket. Units: millimetres, newtons, MPa. Material: Ti-6Al-4V (E = 113.8 GPa, nu
= 0.342, density 4.43 g/cm^3, yield strength 903 MPa). The bracket is bolted to a flat mounting surface, the
plane z = 0, with nothing below it, and carries a clevis for a pin.
Interfaces: (a) four vertical (+z) bolt holes of diameter 10.3124 mm (0.406 in) at (x, y) = (0.0000, 0.0000);
(52.0472, 1.5805); (0.0000, -148.1248); (38.0833, -146.9684); each hole wall must start within 1.0 mm of z = 0
and be at least 5.5 mm long; the bolts are rigid and fully fixed. (b) a clevis of two lugs, each with a bore of
diameter 19.1008 mm on the pin axis through (-20.9236, -74.6978, 44.7246) with direction (-0.030352, 0.999539,
0); measured along the axis from that point, the lug bores span 11.112 to 17.463 mm and -17.463 to -11.113 mm;
the gap between the lugs must stay empty.
Design envelope: x -41 to 72, y -168 to 22, z 0 to 65 mm. Minimum wall / feature thickness 1.27 mm (at most 2 % of
the surface may be thinner). One solid.
Loads, four separate linear static cases, each acting at the point (-21.0359, -75.0647, 44.8010) and spread ...
```

Flags: Original published displacement targets only; source-model reproduction and tolerance remain provisional.; No independent FEM reproduction in this release; end-to-end pass requires evaluator-owned workflow review.

AgentFEM double-arm support brackets (publisher test split)





Aluminium bracket (E 70 GPa), foot undersides fixed, uniform downward seat traction of 1000 or 10000 N. Outputs: mean seat deflection and maximum displacement (publisher summary, 10 %; publisher mesh not converged). The source STEP declared mm but held metre values; the shipped copy is scaled to true mm and checked against the FEM nodes.

Tasks: 5. **Licence:** CC BY 4.0. **Source:** AgentFEM Bracket Elasticity (Haoming Luo, 2026).

Example A-AGENTFEM-fresh_test_000 (hidden references; 14 evidence files):

output	unit	reference	tolerance	class	evidence / provenance
seat_mean_displacement_mm	mm	0.0983548	rel 0.1	published_table	publisher cases/fresh_test_000/summary.json
max_displacement_mm	mm	0.175628	rel 0.1	published_table	publisher cases/fresh_test_000/summary.json

Example prompt (A-AGENTFEM-fresh_test_000), first lines:

```
Support bracket (one member of a parametric double-arm bracket family). geometry.step holds one solid in
millimetres (height 189 mm).

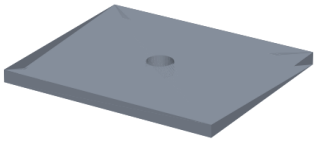
Material: linear isotropic elastic, E = 70 GPa, Poisson ratio 0.33 (density 2700 kg/m^3, gravity not applied).
Supports: the foot undersides, i.e. every face lying in the plane z = 0, are fully fixed (all displacement
components zero).
Load: a uniform downward traction of 1.98728 MPa in the -z direction over the top seat face (the face in the
plane z = 189 mm); total 10000 N.
Analysis: 3-D linear elasticity, small displacements. The publisher used quadratic tetrahedra with a nominal 9 mm
mesh.

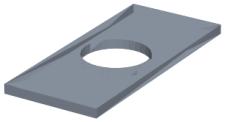
Outputs:
- seat_mean_displacement_mm [mm]: mean of the downward displacement -u_z over the top seat face (z = 189 mm)
- max_displacement_mm [mm]: maximum displacement magnitude anywhere in the bracket

Submit results.json containing exactly an outputs object with these numeric keys. Also submit main.py, part.step,
mesh.msh, fields.npz, solver.log, refinement.json and environment.json. See candidate README for the artifact
contract. Correct numerical answers alone do not establish workflow completion. ...
```

Flags: source STEP unit error fixed: shipped copy scaled x1000 to true millimetres and checked against the FEM nodes; one parametric family: report as a family macro-average; Independently reproduced in this release (evaluator evidence); an end-to-end pass still requires evaluator-owned workflow review.; Independently reproduced (2026-10-14 follow-up, 5 meshes): max displacement within 0.006 %, seat mean within 0.47 % of the published values.

GeoPINONet holed steel plates (validation split)





Structural steel, one end fixed, 1000 N on the opposite end face: compression (-y) and lateral (+x). Outputs: maximum displacement per case from the published ANSYS nodal CSVs (5 %). The archive's fixed/load STL names are swapped relative to the fields; supports and loads are stated from the verified fields.

Tasks: 5. **Licence:** CC BY 4.0. **Source:** GeoPINONet holed-plate FEM dataset (Urrutia Valdes, La Fe-Perdomo, Rodriguez Soto).

Example A-GEOPINO-PLATE-1 (hidden references; 4 evidence files):

output	unit	reference	tolerance	class	evidence / provenance
max_displacement_compression_mm	mm	0.000439557	rel 0.05	published_raw_field	max |u| over the published ANSYS nodal CSV (1_compression.csv / _lateral_bending.csv)
max_displacement_lateral_mm	mm	0.00303832	rel 0.05	published_raw_field	max |u| over the published ANSYS nodal CSV (1_compression.csv / _lateral_bending.csv)

Example prompt (A-GEOPINO-PLATE-1), first lines:

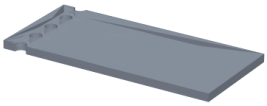
```
Holed steel plate. part.step is one solid in millimetres: a rectangular plate 176.914 mm (x) by 148.475 mm (y) by
10.000 mm (z) with a through hole along z.
Material: ANSYS default Structural Steel, linear isotropic elastic, E = 200 GPa, Poisson ratio 0.3. Linear
static, small displacements, no gravity.
Supports: the end face at y = -74.2373 mm is fully fixed (all displacement components zero).
Two separate load cases, each a total force of 1000 N applied as a uniform traction over the opposite end face (y
= 74.2373 mm):
(1) compression: 1000 N along -y (towards the fixed end);
(2) lateral bending: 1000 N along +x (in-plane shear of the free end).

Outputs:
- max_displacement_compression_mm [mm]: maximum displacement magnitude, load case 1
- max_displacement_lateral_mm [mm]: maximum displacement magnitude, load case 2

Submit results.json containing exactly an outputs object with these numeric keys. Also submit main.py, part.step,
mesh.msh, fields.npz, solver.log, refinement.json and environment.json. See candidate README for the artifact
contract. Correct numerical answers alone do not establish workflow completion. ...
```

Flags: source errors not carried over: BCs stated from the verified fields (archive STL names are swapped), thickness from the CAD (README says 20 mm, CAD and FEM 10 mm); No independent FEM reproduction in this release; end-to-end pass requires evaluator-owned workflow review.

Active-camber morphing-wing cantilever plates (modal)





FDM-printed PLA plates, uniform and with a 1 mm flexible strip, clamped at the end of a 19 mm grip. Outputs: first bending, first torsion and second bending frequencies from the publisher workbook (Abaqus thick shell / Nastran plate models, 5 %). The references are shell idealisations of solid CAD.

Tasks: 5. **Licence:** CC BY 4.0. **Source:** Experimental and numerical data of candidate structural models for active camber morphing wings (Huang & Zhang).

Example A-MORPHWING-Test1 (hidden references; 46 evidence files):

output	unit	reference	tolerance	class	evidence / provenance
f_1B_Hz	Hz	109.05	rel 0.05	published_table	'1-1 Modal Test Experiment and Numerical Results.xlsx' cell S4 (Abaqus thick shell S8R)
f_1T_Hz	Hz	416.67	rel 0.05	published_table	'1-1 Modal Test Experiment and Numerical Results.xlsx' cell S5 (Abaqus thick shell S8R)
f_2B_Hz	Hz	672.94	rel 0.05	published_table	'1-1 Modal Test Experiment and Numerical Results.xlsx' cell S6 (Abaqus thick shell S8R)

Example prompt (A-MORPHWING-Test1), first lines:

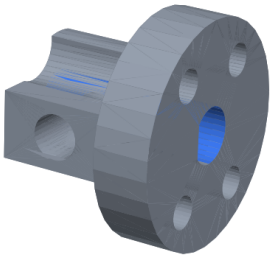
```
Cantilever plate specimen for an active-camber morphing-wing study (FDM-printed PLA): a uniform 4 mm plate.
part.step is one solid in millimetres, 109 x 50 x 4 mm, with a 19 mm clamping grip at x = 0-19 mm (the scalloped
clamp region).
Model the vibrating structure only: the free part x >= 19 mm, clamped (all displacements and rotations zero)
along its whole cross-section at x = 19 mm. The grip is held in the fixture and is not part of the model.
Material: PLA treated as isotropic linear elastic, E = 2107.5 MPa, Poisson ratio 0.3, density 1180 kg/m^3.
Analysis: undamped natural frequencies (modal analysis), no prestress, no gravity. Identify the modes by their
shapes.

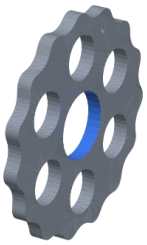
Outputs:
- f_1B_Hz [Hz]: first out-of-plane bending mode
- f_1T_Hz [Hz]: first torsion mode about the span (x) axis
- f_2B_Hz [Hz]: second out-of-plane bending mode

Submit results.json containing exactly an outputs object with these numeric keys. Also submit main.py, part.step,
mesh.msh, fields.npz, solver.log, refinement.json and environment.json. See candidate README for the artifact
contract. Correct numerical answers alone do not establish workflow completion. ...
```

Flags: reference is a shell-model solution of the free length; solid models converge to it within about 1-2 %; No independent FEM reproduction in this release; end-to-end pass requires evaluator-owned workflow review.

3D-printable cycloidal gearbox: component static analyses





FreeCAD/CalculiX studies of 13 printed PETG-CF parts and 2 steel pins, each analysed alone (no gearbox contact). Blue: fixed faces; red: loaded faces (uniform traction, resultant from the source-corrected deck). Output: maximum displacement from the saved FRD files, with a tolerance set by an independent convergence study (see the study page); stress not graded (source error estimate 11-49 %).

Tasks: 5. **Licence:** No repository license; FCStd metadata 'All rights reserved'. Local research only. **Source:** 3D-printable cycloidal drive FEA (jagathsrujan/3d-printable-cycloidal-drive).

Example A-CYC-Coupler-PETGCF (hidden references; 15 evidence files):

output	unit	reference	tolerance	class	evidence / provenance
max_displacement_mm	mm	0.00761747	rel 0.06	published_raw_field	fem/petg_cf/FEM_Coupler_PETGCF/Coupler_fixed.frd (saved CalculiX FRD of the source-corrected deck)

Example prompt (A-CYC-Coupler-PETGCF), first lines:

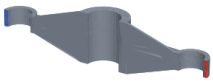
```
Component analysis: the output coupler of an open-source, 3D-printable 15:1 cycloidal gearbox. part.step holds
the single solid in millimetres (33 faces, volume 2063 mm^3). faces.json lists every face with its index, type,
area, centroid and normal/axis; use the geometry, not a CAD system's own face numbering.

Material: PETG-CF, linear isotropic elastic, E = 3000 MPa, Poisson ratio 0.36.
Analysis: linear static, small displacements, no gravity, no contact.
Supports: all three displacement components are zero on face 5: cylinder, area 33.52 mm^2, centroid (76.619,
-0.908, 22.737) mm, radius 2.5 mm, axis (1.0, 0.0, -0.0) through (79.025, -0.0, 20.5) (hole/bore surface); face
6: cylinder, area 117.81 mm^2, centroid (76.525, 1.588, 20.397) mm, radius 2.5 mm, axis (1.0, 0.0, -0.0) through
(79.025, -0.0, 20.5) (hole/bore surface).
Load: total force (100.052, 0.000, 0.000) N applied as a uniform surface traction (constant force per unit area)
over the loaded face(s): face 14 carries (100.052, 0.000, 0.000) N.
For reference, the source solved this with second-order tetrahedra (Netgen, maximum element size 2 mm).

Outputs:
- max_displacemen ...
```

Flags: source author corrected FreeCAD CLOAD units by 1000x; loads stated from the corrected deck; Independently reproduced in this release (evaluator evidence); an end-to-end pass still requires evaluator-owned workflow review.; Tolerance 6 % from an independent convergence study: finest independent 0.00790579 mm vs published 0.00761747 mm (3.8 %), last refinement change 0.87 %.

CADWorld FreeCAD FEM task (publisher mesh protocol)



Publisher geometry, material, constraints and mesh protocol (Gmsh, first-order tetrahedra); references are the values the publisher's evaluator checks (5 %). The publisher's 'Tresca' column is maximum shear stress (s1 - s3)/2 and is now requested under that name. The task reproduces the source's linear model (31 mm deflection of a 260 mm part, stresses above yield); it is not a strength assessment.

Tasks: 1. **Licence:** Apache-2.0 (publisher HF metadata). **Source:** CADWorld FEM benchmark (Zdong104/CADWORLD; HF Zihan1004/CADWORLD).

Example A-CADWORLD-001 (hidden references; 8 evidence files):

output	unit	reference	tolerance	class	evidence / provenance
max_displacement_mm	mm	31.145	rel 0.05	publisher_evaluator	CADWorld reference CSV evaluation_examples/fixtures/fem/freecad-fem-001.csv column 'Displacement Magnitude' ma
max_von_mises_MPa	MPa	705.453	rel 0.05	publisher_evaluator	CADWorld reference CSV evaluation_examples/fixtures/fem/freecad-fem-001.csv column 'von Mises Stress' max (pub
max_shear_stress_MPa	MPa	397.017	rel 0.05	publisher_evaluator	CADWorld reference CSV evaluation_examples/fixtures/fem/freecad-fem-001.csv column 'Tresca Stress' max (publis
max_principal_MPa	MPa	863.519	rel 0.05	publisher_evaluator	CADWorld reference CSV evaluation_examples/fixtures/fem/freecad-fem-001.csv column 'Major Principal Stress' ma

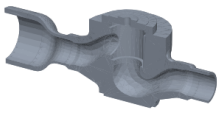
Example prompt (A-CADWORLD-001), first lines:

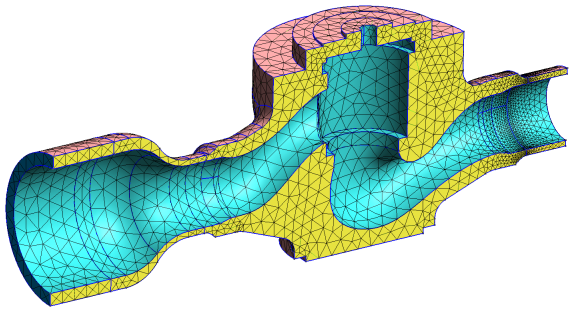
```
FreeCAD/CalculiX benchmark part (Multi-feature aluminium part, end load). part.step is the single solid in
millimetres (38 faces, bounding box [-130.0, -0.0, -30.0, 130.0, 30.0, 30.0] mm). faces.json describes every
face.

Material: AlMg3F24, linear isotropic elastic, E = 70000 MPa, Poisson ratio 0.3.
Analysis: linear static, small displacements, no gravity.
Supports: fixed (all displacement components zero) on face 6: plane, area 100.00 mm^2, centroid (-127.5, 0.0,
0.0) mm, outward normal (-0.0, -1.0, -0.0).
Load: total force 200 N along (0, 1, 0) applied uniformly over face 38: plane, area 100.00 mm^2, centroid (127.5,
-0.0, 0.0) mm, outward normal (-0.0, -1.0, -0.0).
Mesh protocol of the publisher's reference solution (results are compared under this protocol): Gmsh 3-D mesh,
first-order (linear) tetrahedra, maximum element size 2.0 mm, minimum 0.5 mm, CalculiX solver. Report nodal
extrema of that solution.
The publisher column named Tresca Stress is maximum shear stress, (sigma1 - sigma3)/2. Report this
half-difference as max_shear_stress_MPa, not the full principal-stress difference. This task reproduces the
source linear model ...
```

Flags: protocol-bound: publisher mesh (first-order tets) fixes the reference; publisher tolerance 5 %; linear-elastic stresses exceed material yield; computational benchmark, not a strength verdict; No independent FEM reproduction in this release; end-to-end pass requires evaluator-owned workflow review.

Power-plant valve: nonlinear transient heat conduction





Carbon-steel nozzle and stainless bodies (4 solids), ASME temperature-dependent properties, internal temperature ramp 40-250-40 C over 1000 s, weak external convection. Outputs: through-wall temperature differences at two stress classification lines (10 %) and outer-wall temperatures at 100 s, from the published FeenoX history.

Tasks: 1. **Licence:** GPL-3.0. **Source:** FeenoX tutorial #3: heat conduction (valve with carbon and stainless steel parts).

Example A-VALVE-transient (hidden references; 11 evidence files):

output	unit	reference	tolerance	class	evidence / provenance
scl2_dT_max_K	K	14.029	rel 0.1	published_table	published FeenoX history valve-1.csv (192 adaptive time samples)
scl2_dT_min_K	K	-15.183	rel 0.1	published_table	published FeenoX history valve-1.csv (192 adaptive time samples)
scl4_dT_max_K	K	26.128	rel 0.1	published_table	published FeenoX history valve-1.csv (192 adaptive time samples)
scl4_dT_min_K	K	-28.159	rel 0.1	published_table	published FeenoX history valve-1.csv (192 adaptive time samples)
scl2_T_outer_100s_C	C	235.837	rel_offset 0.05	published_table	published FeenoX history valve-1.csv (192 adaptive time samples)
scl4_T_outer_100s_C	C	223.738	rel_offset 0.05	published_table	published FeenoX history valve-1.csv (192 adaptive time samples)

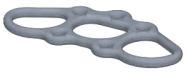
Example prompt (A-VALVE-transient), first lines:

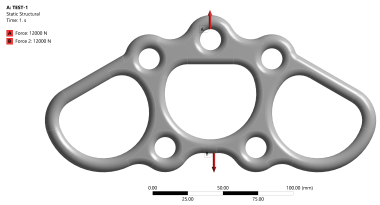
```
Nonlinear transient heat conduction in a power-plant valve: a carbon-steel nozzle welded to stainless-steel valve
bodies (half model, symmetry plane y = 0). valve.brep holds 4 solids in millimetres (112 faces). Solid 2 (the
nozzle, x from 0 to 65 mm) is carbon steel; solids 1, 3 and 4 are stainless steel SA-312. faces.json lists every
face with its index and its boundary group; valve-surfaces.png and valve-materials.png are the source's figures.

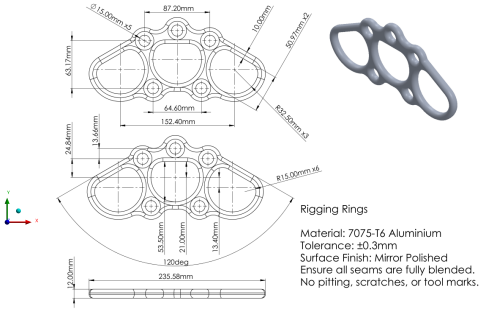
Materials: temperature-dependent conductivity k(T) and thermal diffusivity kappa(T) from asme-properties.fee
(ASME II Part D tables, T in deg C): carbon steel uses k_carbon and kappa_carbon, stainless uses k_312 and
kappa_312. As in the source, k from the tables is in W/(m K) and is multiplied by 1e-3 to W/(mm K); kappa is used
as tabulated (mm^2/s). The volumetric heat capacity follows as k/kappa.
Boundary conditions: 'internal' faces: prescribed temperature T = Tint(t) from internal-temperature.csv (linear
interpolation: 40 C at t = 0, 250 C from 100 s to 500 s, back to 40 C at 600 s, held to 1000 s). 'external' faces:
convection h = 1e-6 W/(mm^2 K) to 50 C. 'symmetry' and both end faces: ...
```

Flags: through-wall differences use the published 192-sample history; 10 % tolerance on differences; temperatures at 100 s graded on the rise above the 40 C initial state; No independent FEM reproduction in this release; end-to-end pass requires evaluator-owned workflow review.

7075-T6 rigging plate: five ANSYS load tests and the drawing-to-CAD task (figure_reference)







Supplied STEP (track A) or the dimensioned manufacturing drawing (track B) plus the source's boundary-condition figure per test. Output: maximum von Mises stress from the original legend (20 %). Track B: the built part must overlap the original CAD within 2 % symmetric difference after alignment. Material card and remote-coupling details are not published; tests 4 and 5 are left out (conflicting load labels; revised TEST-5R results).

Tasks: 5. **Licence:** CERN-OHL-S-2.0. **Source:** OpenRiggingPlate (mpangch).

Example A-RIGGING-test1 (hidden references; 5 evidence files):

output	unit	reference	tolerance	class	evidence / provenance
max_von_mises_MPa	MPa	526.74	rel 0.2	published_legend	FEA Analysis/Test 1 (2697.7 lbf)/Stress.png legend '526.74 Max'

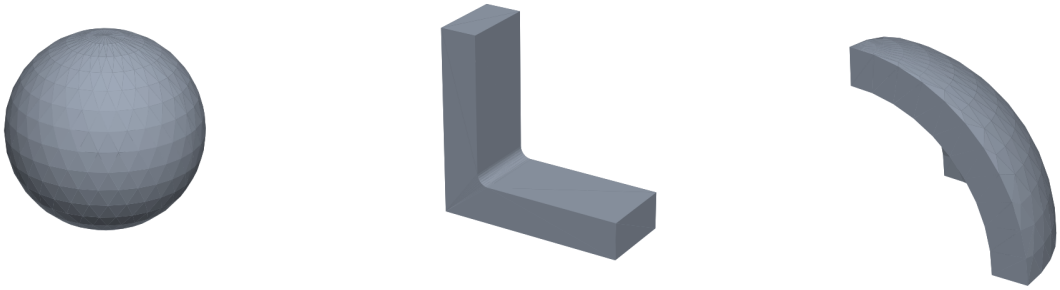
Example prompt (A-RIGGING-test1), first lines:

```
Hard-anodised 7075-T6 rigging plate, load test 1 of the source's seven ANSYS load cases (Test 1 (2697.7 lbf)).
riggingplate.step is the single solid in millimetres (235.58 x 111.90 x 12 mm, 145 faces).
Material: aluminium 7075-T6, linear isotropic elastic (standard handbook values; the source did not publish its
card, and the graded stress does not depend on Young's modulus for these loads).
Analysis: linear static, small displacements (ANSYS Mechanical in the source).
Boundary conditions: exactly as shown in boundary-conditions.png (the source's own ANSYS figure): each labelled
scope (A, B, ...) is a face or set of faces of the plate; 'Force' applies the stated total force uniformly
distributed over the scoped faces in the drawn direction; 'Remote Displacement' fixes the scoped faces through a
rigid remote point (all displacements and rotations zero). Where only two equal and opposite forces are applied
and no support is drawn, the plate is self-equilibrated; remove rigid-body motion without adding load (e.g. weak
springs or inertia relief), as the source did.
Legend of this test's figure: A Force: 12000 N; B; Force 2: 12000 N.

Ou ...
```

Flags: load scopes given by the source figure, not by face IDs; reference is a rounded legend maximum; peak at load-face edges is mesh-sensitive (20 % tolerance); No independent FEM reproduction in this release; end-to-end pass requires evaluator-owned workflow review.

FEMCAD-Hard: specification-to-CAD-to-FEM tasks (track B)



Five tasks from the Wolfram FEM documentation: spherical capacitor shell (electrostatics), L-bracket with a filleted inner corner, octant of a spherical pressure hull, plate with a hole in tension and a thick tube heated on an inner band. The model reads the written specification, builds the solid, meshes and solves it. References are closed-form or page-stated values with the page's tolerance; the built part is gated against the reference solid.

Tasks: 5. **Licence:** Wolfram documentation (copyright Wolfram Research); figures not redistributable. **Source:** Wolfram FEM documentation: Spherical Capacitor.

Example B-FEMCAD-057 (hidden references; 5 evidence files):

output	unit	reference	tolerance	class	evidence / provenance
max_von_mises_MPa	MPa	230.197	rel 0.05	published_table	source page; https://reference.wolfram.com/language/PDEModels/tutorial/StructuralMechanics/SolidMechanics.html

Example prompt (B-FEMCAD-057), first lines:

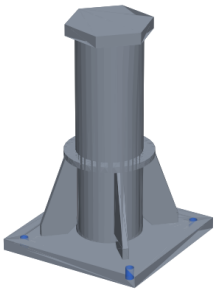
```
CAD + FEM task. Build the part with a CAD script and write it as part.step in millimetres in exactly the
coordinate frame given (multi-region models: one STEP with one solid per region and shared, coincident interface
faces; 2-D models: planar faces in the z = 0 plane); then analyse YOUR model with any open-source FEM library,
set up exactly as under Analysis. main.py must run both steps (artifact contract below). Report mesh-converged
values (each changes by less than 1 % under refinement). Your results are compared with the reference analysis.

Geometry. Union of the boxes x in [0, 10], y in [0, 20], z in [0, 50] (upright) and x in [0, 50], y in [0, 20], z
in [0, 10] (foot), units mm; fillet of radius 3 on the inner corner edge (x = 10, z = 10, along y).

Analysis. E = 210000 MPa, Poisson ratio 0.3. Top face z = 50 fixed; 1000 N in -z as a uniform traction (5 MPa) on
the end face x = 50. Linear elastic.
The evaluator aligns part.step to a reference solid built from this specification (bounding-box centres, the 24
proper axis rotations; mirror images are not accepted) and requires a relative symmetric volume difference of at
most 2 ...
```

Flags: geometry check pending: reference CAD from the source construction; No independent FEM reproduction in this release; end-to-end pass requires evaluator-owned workflow review.

fcVM plastic collapse (limit) load factors (v1.5)



Steel parts from the fcVM workbench analysed as elastic-perfectly plastic von Mises bodies under proportional loading (small strain). Blue: supported faces; red: loaded faces. Output: the collapse load factor of the reference load set, graded against fcVM's published plateau value with a tolerance set by an independent limit-analysis convergence study (see the v1.5 study page).

Tasks: 1. **Licence:** LGPL-2.1-or-later (repository); example data licence not stated separately. Local research use. **Source:** fcVM workbench examples (Harry van Langen, HarryvL/fcVM-workbench).

Example A-FCVM-PL-Wyk_Rohrstoss_ring (hidden references; 14 evidence files):

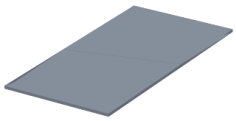
output	unit	reference	tolerance	class	evidence / provenance
limit_load_factor	-	1.52	rel 0.05	published_table	fcVM output files/Wyk_Rohrstoss_ring.out 'load' column: maximum, on a flat plateau (26 printed rows within 1 %

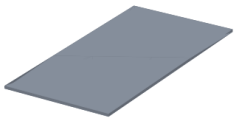
Example prompt (A-FCVM-PL-Wyk_Rohrstoss_ring), first lines:

```
Plastic collapse (limit load) analysis of a steel part from the fcVM workbench examples (Wyk_Rohrstoss_ring).
analysis.step holds the analysis geometry in millimetres (12 solid region(s) that share their interfaces and act
as one continuous body, volume 2577880 mm^3). faces.json describes every face (index, type, area, centroid,
normal/axis).
Material: elastic-perfectly plastic von Mises (associated flow, no hardening), E = 210000 MPa, Poisson ratio 0.3,
yield stress 235 MPa. Small strains and small displacements (geometrically linear).
Supports: (1) all displacement components = 0 on face 5: cylinder, area 879.65 mm^2, centroid (105.0, -105.0,
10.0) mm, radius 7.0 mm, axis (-0.0, -0.0, -1.0) through (105.0, -105.0, 0.0) (boss/shaft surface); face 6:
cylinder, area 879.65 mm^2, centroid (-105.0, -105.0, 10.0) mm, radius 7.0 mm, axis (-0.0, -0.0, -1.0) through
(-105.0, -105.0, 0.0) (boss/shaft surface); face 7: cylinder, area 879.65 mm^2, centroid (105.0, 105.0, 10.0) mm,
radius 7.0 mm, axis (-0.0, -0.0, -1.0) through (105.0, 105.0, 0.0) (boss/shaft surface); face 8: cylinder, area
879.65 mm^2, centroid (-105.0, 105.0, 10.0) mm, radi ...
```

Flags: load set matches the published fcVM load sums (max relative difference 1.2e-05); independent finest-mesh limit factor 1.524 vs published 1.52 (+0.2 %), last refinement change 0.03 %; Independently reproduced in this release (evaluator evidence); an end-to-end pass still requires evaluator-owned workflow review.

Code_Aster SSLP311: crack on an inclined bimaterial interface (v1.5)





A 4 x 8 mm plate of two elastic materials (E = 2.0e6 / 2.0e5 MPa) bonded along an inclined line through the centre, with a 2 mm crack on that line, in plane-stress tension. Supplied CAD: the domain extruded 0.1 mm, two solids, crack faces split out. Outputs: the energy release rate at both tips, graded against the published boundary-element solution with a tolerance from an independent J-integral study.

Tasks: 4. **Licence:** Code_Aster sources GPL-3.0 (see LICENSE-Code_Aster); documentation (c) EDF R&D. Local research use. **Source:** Code_Aster validation test SSLP311 (EDF R&D), reference solution Miyazaki, Ikeda, Soda & Munakata, Eng. Fract. Mech. 45 (1993) 599-610.

Example A-ASTER-SSLP311-theta15 (hidden references; 21 evidence files):

output	unit	reference	tolerance	class	evidence / provenance
G_right_tip	J/m^2	101.251	rel 0.05	published_table	sslp311a.comm VALE_REFE (FOND1, right tip, theta = 15 deg)
G_left_tip	J/m^2	96.7362	rel 0.05	published_table	sslp311a.comm VALE_REFE (left tip, theta = 15 deg)

Example prompt (A-ASTER-SSLP311-theta15), first lines:

```
Linear-elastic fracture analysis of a central crack lying on an inclined bimaterial interface in a finite plate
under uniform tension (Code_Aster validation test SSLP311). plate.step holds the plate in millimetres as two
solids: solid 1 (material 1) above the interface line and solid 2 (material 2) below it. The 2-D domain is
extruded 0.1 mm along +z from z = 0 only as an exchange convention; analyse the z = 0 cross-section as a 2-D
plane-stress problem. faces.json describes every face (index, type, area, centroid, normal).
Geometry: the plate occupies x from -2 to 2 mm and y from -4 to 4 mm. The interface is the straight line through
the origin at 15 degrees to the +x axis. The crack is the 2 mm long segment of that line centred on the origin,
with tips at (0.965926, 0.258819) mm (right tip) and (-0.965926, -0.258819) mm (left tip). Its faces (face 2 of
solid 1 and face 12 of solid 2) are traction-free and not connected; the rest of the interface is perfectly
bonded.
Materials: solid 1 E = 2.0e6 MPa, Poisson ratio 0.3; solid 2 E = 2.0e5 MPa, Poisson ratio 0.3. Linear elastic,
small strain.
Loads and supports: uniform traction sigma ...
```

Flags: independent finest-mesh G: right 101.2 (-0.07 %), left 96.62 (-0.12 %) J/m^2; last level change 0.07 %; the source's own finite-element run (theta = 15 deg) is within 0.3 % of the same reference; Independently reproduced in this release (evaluator evidence); an end-to-end pass still requires evaluator-owned workflow review.

fcVM elastic buckling of steel members and welded assemblies (v1.3)



FreeCAD models from the fcVM workbench (Harry van Langen), with face supports, edge supports, distributed forces, pressures and self-weight as in the original constraints. Blue: supported faces; red: loaded faces. Output: first elastic buckling factor of the reference load set, graded against the factor in the published fcVM output, with a tolerance set by an independent eigenvalue convergence study (see the study page).

Tasks: 4. **Licence:** LGPL-2.1-or-later (repository); example data licence not stated separately. Local research use. **Source:** fcVM workbench examples (Harry van Langen, HarryvL/fcVM-workbench).

Example A-FCVM-buckling_U100_3D (hidden references; 16 evidence files):

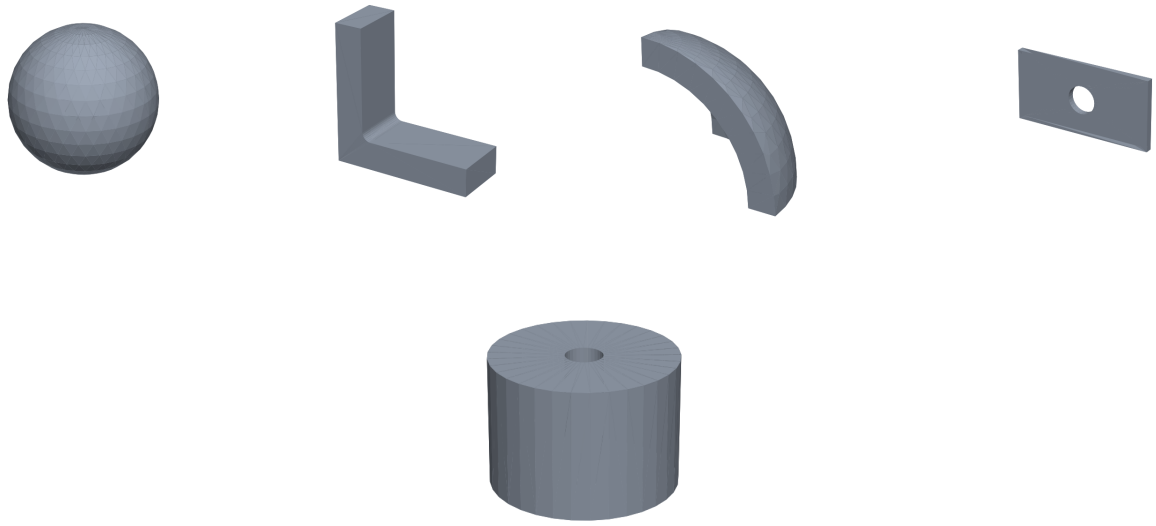
output	unit	reference	tolerance	class	evidence / provenance
lambda1	-	3.89514	rel 0.05	published_table	fcVM output files/buckling_U100_3D.out header 'elastic buckling factors'

Example prompt (A-FCVM-buckling_U100_3D), first lines:

```
Linear (eigenvalue) buckling analysis of a steel structure from the fcVM workbench examples (buckling_U100_3D).
analysis.step holds the analysis geometry in millimetres (1 solid region(s) that share their interfaces and act
as one continuous body, volume 1207500 mm^3). faces.json describes every face (index, type, area, centroid,
normal/axis).
Material: linear isotropic elastic, E = 210000 MPa, Poisson ratio 0.3.
Supports: (1) all displacement components = 0 on face 11: plane, area 1207.50 mm^2, centroid (19.502, 0.0, 0.0)
mm, outward normal (-0.0, -0.0, -1.0).
Reference load set (load factor 1): (1) uniform traction (-0.0, -0.0, -41.407867) MPa on face 10: plane, area
1207.50 mm^2, centroid (19.502, 0.0, 1000.0) mm, outward normal (0.0, 0.0, 1.0). (2) uniform traction (1.656315,
0.0, 0.0) MPa on face 10: plane, area 1207.50 mm^2, centroid (19.502, 0.0, 1000.0) mm, outward normal (0.0, 0.0,
1.0). Self-weight is part of the reference load: density 7900 kg/m^3, gravity (0.0, 0.0, -10.0) m/s^2 (body force
(0.0, 0.0, -7.9e-05) N/mm^3).
Analysis: compute the linear prebuckling stress state under the reference load set, then the elastic b ...
```

Flags: load set matches the published fcVM load sums (max relative difference 8.9e-05); independent finest-mesh factor 3.8606 vs published 3.89514 (-0.9 %), last refinement change 0.01 %; Independently reproduced in this release (evaluator evidence); an end-to-end pass still requires evaluator-owned workflow review.

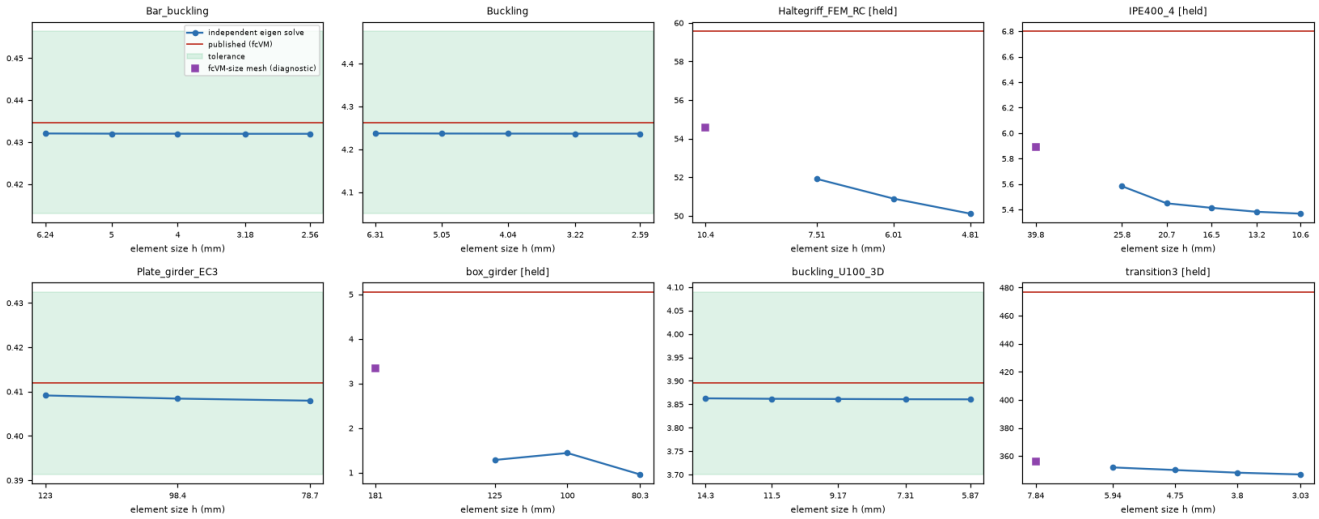
Geometry gates for drawing/specification-to-CAD tasks (v1.3)



task	reference solid	volume (mm^3)	volume check	rotated + moved	scaled x1.03
B-FEMCAD-031	spherical shell r 3-10 mm	4075.69	3e-08	0	rejected: analytic sphere/cylinder radius mismatch
B-FEMCAD-057	L-bracket, R3 concave fillet on the inner corner edge	18038.6	2e-08	0	rejected: analytic sphere/cylinder radius mismatch
B-FEMCAD-058	octant of a hollow sphere r 80-100 mm	255516	2e-09	0	rejected: analytic sphere/cylinder radius mismatch
B-FEMCAD-062	plate with a through hole r 10 mm	23429.2	1e-08	2.2e-15	rejected: analytic sphere/cylinder radius mismatch
B-FEMCAD-067	tube r 20-100 mm, length 140 mm	4.2223e+06	1e-10	6.6e-14	rejected: analytic sphere/cylinder radius mismatch
B-RIGGING-test1	original source CAD (rigging plate)			0	0.246

Order of the renders: the table rows. Reference solids are built from the task text (FEMCAD-Hard pages, NAFEMS LE1/LE11 geometry definitions) or taken from the source CAD (LE10). Each is accepted only if its OpenCASCADE volume agrees with a closed form, or with Pappus's theorem for the revolved LE11 profile, to 1e-4. The scorer aligns the submitted part.step to the reference using bounding-box centres and the 24 proper axis rotations (mirror images are rejected), and passes it if the symmetric volume difference is at most 2 % of the reference volume. Probe columns: symmetric difference for the reference rotated 90 degrees and moved (must pass) and for it scaled by 3 % (must fail). The gate is a necessary check; features, regions and boundary-condition faces are covered by the trusted review.

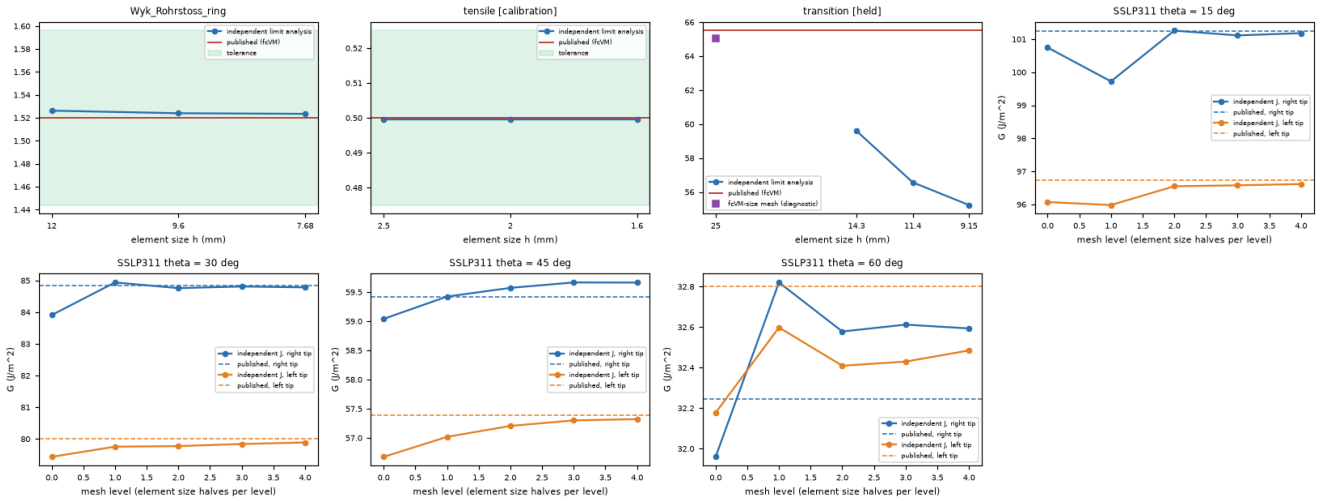
fcVM elastic buckling: correspondence and independent eigenvalue study (v1.3)



model	published lambda1	meshes	finest h (mm)	finest independent	difference	last change	decision
Bar_buckling	0.43472	5	2.56	0.43199	-0.6 %	0.00 %	admitted, tol 5 %
Buckling	4.26305	5	2.59	4.2366	-0.6 %	0.00 %	admitted, tol 5 %
Haltegriff_FEM_RC	59.5774	3	4.81	50.115	-15.9 %	1.55 %	held
IPE400_4	6.80163	5	10.6	5.3673	-21.1 %	0.27 %	held
Plate_girder_EC3	0.41191	3	78.7	0.40796	-1.0 %	0.11 %	admitted, tol 5 %
box_girder	5.05593	3	80.3	0.95927	-81.0 %	50.16 %	held
buckling_U100_3D	3.89514	5	5.87	3.8606	-0.9 %	0.01 %	admitted, tol 5 %
transition3	476.994	4	3.03	346.83	-27.3 %	0.36 %	held

Correspondence: scripts/cadfem_bench/fcvm_extract.py reads each original FreeCAD document as data, without executing it. It maps every constraint's references onto the analysis solid and rebuilds the reference load the way fcVM's own code does: the force direction vector over the total face area, signed pressure along the outward normal, and self-weight from density times the control-file gravity. All 14 fcVM models reproduce the 'Sum of loads' printed in their published output; the largest difference is 5.8e-4, from a value printed to 3 significant figures. Independent study: the analysis solid is remeshed with gmsh (solids fragmented into one conforming body) on quadratic tetrahedra, with element size reduced by 0.8 per step. Where finer meshes exceed the study machine's memory, the sequence is completed on the coarse side with the same ratio. A FEniCSx prebuckling solve is followed by the geometric stiffness K_G, and SLEPc solves (-K_G) x = mu K x for the largest mu (lambda = 1/mu, so load-reversal modes do not interfere). The bar case agrees with Euler's cantilever value to 0.05 %. Admission uses the cycloidal rule: last change < 1 %, previous < 2 %, band = |finest - published| / published + 2 x last change <= 10 %, tolerance = band (at least 5 %). Purple squares: a diagnostic solve on a coarser mesh near fcVM's element count, shown where the gap is large; it never changes a decision. Held: Haltegriff_FEM_RC: independent buckling factor not settled (last change 1.55 %, previous 2.01 %; finest 50.115, -15.9 % from the published factor 59.5774); a diagnostic solve on a coarser mesh (24728 elements; fcVM used 16564) gives 54.57: part of the gap closes on coarsening, so the published value is at least partly a coarse-mesh value; IPE400_4: settled independent value 5.3673 is -21.1 % from the published factor 6.80163 (band 22 % > 10 %); a diagnostic solve on a coarser mesh (8108 elements; fcVM used 4900) gives 5.889: part of the gap closes on coarsening, so the published value is at least partly a coarse-mesh value; box_girder: independent buckling factor not settled (last change 50.16 %, previous 10.86 %; finest 0.95927, -81.0 % from the published factor 5.05593); a diagnostic solve on a coarser mesh (27664 elements; fcVM used 11694) gives 3.344, but the independent factor is non-monotonic under refinement (strongly mesh-dependent), so no cause is assigned; transition3: settled independent value 346.83 is -27.3 % from the published factor 476.994 (band 28 % > 10 %); a diagnostic solve on a coarser mesh (16707 elements; fcVM used 12864) gives 355.9: coarsening barely moves it, so mesh size does not explain the gap and the set-up correspondence beyond the load sum is unresolved.

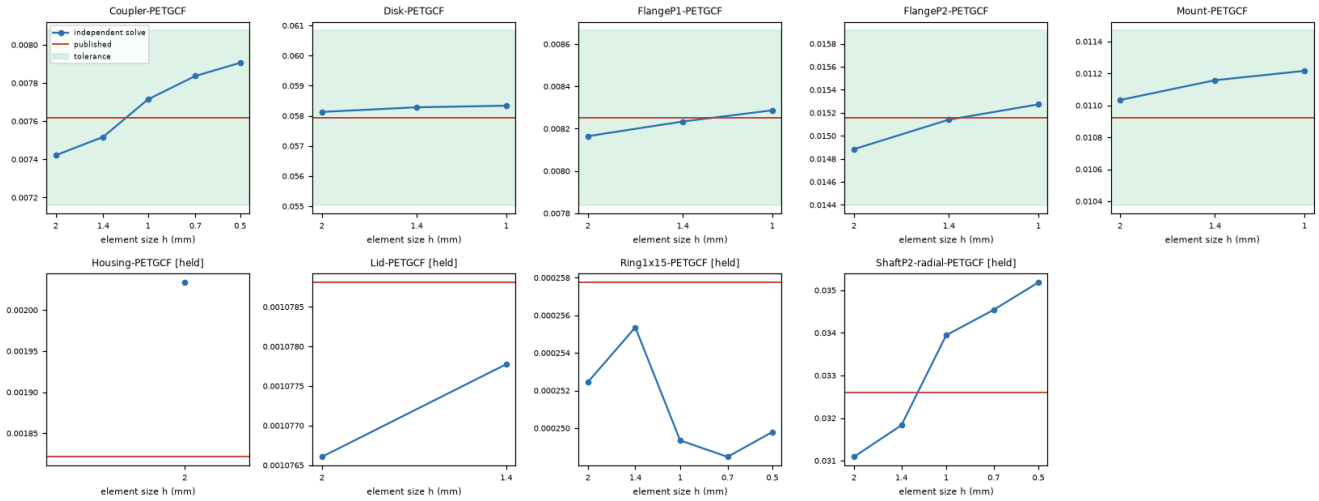
v1.5 studies: plastic limit loads and interface-crack energy release rates



task	output	published	finest independent	difference	last change	decision
A-ASTER-SSLP311-theta15	G_right_tip	101.2507	101.1823	-0.07 %	0.07 %	primary, tol 5 %
A-ASTER-SSLP311-theta15	G_left_tip	96.7362	96.6164	-0.12 %	0.04 %	primary, tol 5 %
A-ASTER-SSLP311-theta30	G_right_tip	84.8415	84.7830	-0.07 %	0.03 %	primary, tol 5 %
A-ASTER-SSLP311-theta30	G_left_tip	80.0173	79.8881	-0.16 %	0.06 %	primary, tol 5 %
A-ASTER-SSLP311-theta45	G_right_tip	59.4124	59.6616	+0.42 %	0.00 %	primary, tol 5 %
A-ASTER-SSLP311-theta45	G_left_tip	57.3827	57.3219	-0.11 %	0.04 %	primary, tol 5 %
A-ASTER-SSLP311-theta60	G_right_tip	32.2438	32.5934	+1.08 %	0.06 %	primary, tol 5 %
A-ASTER-SSLP311-theta60	G_left_tip	32.8017	32.4847	-0.97 %	0.17 %	primary, tol 5 %
A-FCVM-PL-Wyk_Rohrstoss_ring	limit_load_factor	1.52	1.524	+0.2 %	0.03 %	primary, tol 5 %
A-FCVM-PL-tensile	limit_load_factor	0.5	0.4994	-0.1 %	0.00 %	calibration, tol 5 %
A-FCVM-PL-transition	limit_load_factor	65.5	55.23	-15.7 %	2.42 %	held

Plastic limit loads: published value = fcVM's maximum load factor, on a flat plateau of the published load-displacement curve. The load set is rebuilt from the FreeCAD constraints and matches the published load sum. Independent check: FEniCSx, quadratic tetrahedra, elastic-perfectly plastic von Mises with radial return and the consistent tangent, proportional loading with bisection; the limit is bracketed to 0.2 %. The solver reproduces the tensile bar's closed-form 0.5. Interface crack: published G from the source (boundary-element solution, stated uncertainty below 0.1 %). Independent check: FEniCSx plane stress, quadratic triangles graded to the tips, equivalent-domain J-integral on three rings, five mesh levels; it reproduces a Griffith crack within 0.1 %. Admission rule as for the other studied families: last change < 1 %, previous < 2 %, band <= 10 %, tolerance = band (at least 5 %).

Cycloidal convergence study (v1.2)



part	published (mm)	finest independent (mm)	difference	meshes	tolerance
Coupler-PETGCF	0.00761747	0.00790579	+3.8 %	5	6 %
Disk-PETGCF	0.0579329	0.058337	+0.7 %	3	5 %
FlangeP1-PETGCF	0.00825116	0.00828686	+0.4 %	3	5 %
FlangeP2-PETGCF	0.0151561	0.015273	+0.8 %	3	5 %
Mount-PETGCF	0.010922	0.0112161	+2.7 %	3	5 %

Independent solves: supplied STEP remeshed with gmsh, second-order geometry, quadratic displacement, FEniCSx/PETSc MUMPS, loads as constant traction normalised to the meshed area. Admitted parts: refinement settled (last change < 1 %, previous < 2 %), force-balance residual < 1e-6 (the study script's 1e-8 cut-off is relaxed because 1e-8 to 1e-7 residuals are exact at these tolerances), band <= 10 %. Tolerance = |finest - published| / published + 2 x last change, rounded up to 1 %, at least 5 %. Held: Housing and Lid (finer meshes exceed the study's 220k-element solver guard), Ring1x15 (residual 4.7e-4, values do not settle), ShaftP2-radial (still changing 1.8 % per refinement, 7.9 % from the published value). The study files are in the evaluator evidence bundle; they never replace the published targets.

Held tasks and source registry

held tasks	reason
40	External design evaluator and environment not packaged; locally defined design variants excluded from published-source evaluation.
2	Source STEP imports but fails OpenCASCADE BRepCheck validity; no silent repair.
1	Mesh-convergence study: only 1 mesh(es) solved; finer meshes exceed the study's 220k-element direct-solver guard, no converged estimate.
1	Mesh-convergence study: only 2 mesh(es) solved; finer meshes exceed the study's 220k-element direct-solver guard, no converged estimate.
1	Mesh-convergence study: force-balance residual 4.7e-04 and refinement values 0.00025245, 0.00025536, 0.00024935, 0.00024847, 0.00024979 mm do not settle cleanly.
1	Mesh-convergence study: not converged: last refinement change 1.83 %, previous 1.72 %; finest independent value 7.9 % from the published value (empirical band 12 %).
1	Independent eigenvalue study: independent buckling factor not settled (last change 1.55 %, previous 2.01 %; finest 50.115, -15.9 % from the published factor 59.5774); a diagnostic solve on a coarser mesh (24728 elements; fcVM used 16564) gives 54.57: part of the gap closes on coarsening, so the published value is at least partly a coarse-mesh value.
1	Independent eigenvalue study: settled independent value 5.3673 is -21.1 % from the published factor 6.80163 (band 22 % > 10 %); a diagnostic solve on a coarser mesh (8108 elements; fcVM used 4900) gives 5.889: part of the gap closes on coarsening, so the published value is at least partly a coarse-mesh value.
1	Independent eigenvalue study: independent buckling factor not settled (last change 50.16 %, previous 10.86 %; finest 0.95927, -81.0 % from the published factor 5.05593); a diagnostic solve on a coarser mesh (27664 elements; fcVM used 11694) gives 3.344, but the independent factor is non-monotonic under refinement (strongly mesh-dependent), so no cause is assigned.
1	Independent eigenvalue study: settled independent value 346.83 is -27.3 % from the published factor 476.994 (band 28 % > 10 %); a diagnostic solve on a coarser mesh (16707 elements; fcVM used 12864) gives 355.9: coarsening barely moves it, so mesh size does not explain the gap and the set-up correspondence beyond the load sum is unresolved.
1	Independent limit-analysis study: independent limit load not settled (last change 2.42 %, previous 5.36 %; finest 55.23, -15.7 % from the published factor 65.5); a diagnostic solve on a coarser mesh (3215 elements; fcVM used 4866) gives 65.04: most of the gap closes on coarsening, consistent with a coarse-mesh published value.

Source registry: 95 collected collections (74 held, 13 pilot, 8 excluded). Largest held pools that could become tasks with further work: Code_Aster validation cases (50 cases, 761 published checks), SFEM (61), GeoPINONet lugs (80), Bath auxetics (18).

collection	status	tasks / candidates	blocker / decision
SimJEB GE jet engine bracket (381 human designs)	pilot	5	Analysis displacement vs SimJEB published OptiStruct; peak stress/verdict and design tasks via the validated benchmark FEM grader. 13 redesign/lighten tasks without a passing human witness design (unproven feasibility) are left out.
AgentFEM bracket elasticity (176 pairs)	pilot	5	Publisher test split (32) admitted with the source STEP unit error fixed (metre-valued coordinates under a mm header; shipped copies scaled to true mm and checked against the FEM nodes); train/validation (144) kept out to avoid redundancy within one parametric family.
GeoPINONet holed plates (80 pairs)	pilot	5	Validation split (16) admitted with BCs stated by geometry (archive fixed/load STL names are swapped relative to the fields).
Morphing-wing cantilever plates (modal)	pilot	5	Tests 1-3, 5-9 admitted; Test 4 excluded (STEP fef50 % vs result block f75 %).
3D-printable cycloidal drive component FEM (31 saved solutions)	pilot	5	15 admitted (PETG-CF set + steel pins), displacement only: the source's CalculiX error estimate at the stress peaks is 11-49 %, so stress is not graded. PLA reruns of the same geometry/loads, the axial-cam variant, the mesh-only housing FRD and the uncorrected disk deck are not separate tasks.
CADWorld FreeCAD/CalculiX FEM tasks (3)	pilot	1	001 admitted; 002 left out (support on a single edge, load on an auxiliary Box object; 76.6 GPa peak is a singular-support artefact); 003 left out (20 MN on a 30 mm specimen gives 811 mm displacement, outside small-displacement analysis).
NAFEMS LE1/LE10/LE11 via FeenoX	pilot	0	Calibration tier (public benchmarks, answers may be recalled); GNL5 large-deflection case not packaged.
FEMCAD-Hard v1 (Wolfram FEM docs)	pilot	5	Five tasks with closed-form/page-stated references admitted.
FeenoX power-plant valve transient thermal	pilot	1	Admitted; Gmsh surface tags verified equal to OpenCASCADE face indices.
OpenRiggingPlate 7075-T6 (7 ANSYS load tests + drawing)	pilot	5	Silver: references are rounded legend maxima; load scopes come from the source figures. Tests 4 (README/folder/figure load labels disagree) and 5 (results labelled TEST-5R, a revised set-up) left out.

collection	status	tasks / candidates	blocker / decision
fcVM workbench examples: elastic buckling (8 FreeCAD models)	pilot	4	First elastic buckling factor from each published fcVM output. The load set rebuilt from the FreeCAD constraints (fcVM's own rules) matches every published 'Sum of loads'. A case is graded only if an independent FEniCSx/SLEPc eigenvalue study settles within 10 % of the published factor; the rest are held with the reason.
fcVM workbench examples: plastic limit loads (3 models whose published curve ends on a flat plateau)	pilot	1	Published fcVM collapse factor, elastic-perfectly plastic von Mises, geometrically linear. Load sets match the published load sums. Graded only if an independent FEniCSx limit analysis settles within the 10 % band; the tensile bar (closed form) is calibration.
Code_Aster SSLP311: central crack on an inclined bimaterial interface (4 angles)	pilot	4	Published boundary-element energy release rates at both tips (Miyazaki et al. 1993, tabulated by the source); geometry built from the V-document; graded only if an independent J-integral study settles within the 10 % band.
COMSOL legacy studies (turbine stator, gears, muffler, ...)	excluded	7	needs a proprietary solver/material model to define the problem - COMSOL-only solved models; quantitative fields require COMSOL export.
Ansys ACP workshops (4)	excluded	4	no FEM result - No saved results.
Piston assembly (duplicate source)	excluded	0	duplicate, superseded or locally generated - Byte-identical duplicate of the piston source.
6-axis robot arm	excluded	1	no FEM result - Rigid-body joint motion only; no deformable FEM.
CAD/drawing leads without FEM output	excluded	7	no FEM result - No saved or published FEM result.
Test-bench strip restarts	excluded	4	duplicate, superseded or locally generated - Duplicate frame CAD; strip results have no matching CAD.
SFEM method re-run on ABC CAD	excluded	489	duplicate, superseded or locally generated - Loads generated locally; the user asked for real engineering set-ups, not invented loads.
Engineering FEM v1	excluded	14	duplicate, superseded or locally generated - Superseded by later collections.
DeepWheel free-free wheel modes (10 downloaded of 904)	held	10	conflicting or unmatched source references - CAD mass at the publisher density (2680 kg/m^3) is 3.25 % below the publisher CSV mass; the density behind the published frequencies cannot be determined.
SFEM stochastic linear elasticity (61)	held	61	no deterministic, mesh-independent numerical reference - Fields are medians over random, unseeded load clumps whose locations were not stored: no deterministic set-up to give a model.
Bath curved-link auxetic lattices (18)	held	18	CAD only in a proprietary native format - CAD is Autodesk Inventor .ipt only; the ANSYS .dat decks carry the mesh, so no open CAD exists for a CAD->FEM task.
Auxetic meta-shells torsional buckling (6)	held	6	supports/loads/materials not recoverable as CAD-face set-up - References are torque-twist curves of nonlinear Abaqus shell models; the shell section, contact and loading must be transcribed from the decks first.
Scan-based cold-formed steel columns (3)	held	3	no B-rep CAD (mesh, scan or paper geometry only) - Geometry is a scan STL only (no B-rep); imperfections are encoded in the mesh.
NAFEMS GNL5 large-deflection beam	held	1	supports/loads/materials not recoverable as CAD-face set-up - Not in the pilot packaging; geometric-nonlinear prompt and load history not yet transcribed.
Code_Aster validation cases (50, 761 published checks)	held	49	no B-rep CAD (mesh, scan or paper geometry only) - No CAD (GIBI/MED meshes; 28 geometry scripts); checks are keyed to mesh groups. Prompts must be transcribed from the French V-documents. SSLP311 is admitted under codeaster-sslp311 (CAD built from the documented geometry). Reviewed and held in v1.5: SSNV104 Hertz spheres (the analytical reference assumes a << R, but a/R = 0.2 here and the source's own result differs by 13.5 %); SSNV506 indentation (approximate textbook plasticity estimate for a 100 mm indentation).
FEMCAD Source v2 (Wolfram, 100 linked cases)	held	100	no deterministic, mesh-independent numerical reference - Task specifications only: no gold answers were extracted (no solver runs by policy).
COMSOL application-gallery joint tasks (50)	held	50	needs a proprietary solver/material model to define the problem - No numerical references or tolerances; source models need COMSOL.
DeepJEB brackets (2,138 scalar results)	held	2138	no B-rep CAD (mesh, scan or paper geometry only) - Only one STEP downloaded (shell, not solid); same GE load set as SimJEB, which is admitted.
FrontISTR FreeCAD benchmarks: gear, can, bike frame, spring	held	4	no deterministic, mesh-independent numerical reference - Only peak von Mises is published and it keeps rising with refinement (gear 334 -> 397 -> 442 MPa): no mesh-independent target.
GeoPINONet 3-D lugs (80)	held	80	supports/loads/materials not recoverable as CAD-face set-up - Pin-bore load distribution (uniform force vs bearing) is not documented, and the archive's fixed/load STL names are swapped.

collection	status	tasks / candidates	blocker / decision
UCSM sheet-metal inverse forming (8)	held	8	supports/loads/materials not recoverable as CAD-face set-up - LS-DYNA inverse one-step strains on surface CAD; material card and process input not supplied.
Tricuspid heart valve (Abaqus/Explicit VUMAT)	held	1	needs a proprietary solver/material model to define the problem - Shell-surface IGES and a user material subroutine; replay needs the published VUMAT inside Abaqus.
fcFEM collapse examples (5)	held	5	conflicting or unmatched source references - Saved snapshots cannot be matched to load factors/increments from the published parameter files.
Aerospace wing box (IGES + APDL)	held	1	conflicting or unmatched source references - IGES faces only; geometry/unit/set-up correspondence with the published results unresolved.
Cemented hip implant contact (Leeds)	held	1	supports/loads/materials not recoverable as CAD-face set-up - Pelvis/cement are open IGES surfaces; only one of four wear states maps to an executable deck.
HIRENASD wing modes (NASA)	held	1	no deterministic, mesh-independent numerical reference - Mode-shape fields only; frequency file not public; frame correspondence unresolved.
Blade dovetail reduced model (PoliTo)	held	1	no deterministic, mesh-independent numerical reference - Reduced mass/stiffness matrices only, no response results.
Fokker F28 fuselage crash (TU Delft)	held	3	no deterministic, mesh-independent numerical reference - Results only as plots mixing FEM and experiment; no machine-readable response.
WAAM wall thermal history and distortion (AIMEN/SCAI)	held	1	supports/loads/materials not recoverable as CAD-face set-up - No solver deck (moving heat source parameters only in PDFs); VMAP unit tags undefined.
Capriccio FE/MD fracture, Digimat cure, bimetal strip, SOFA actuator, tapered spring	held	6	needs a proprietary solver/material model to define the problem - Coupled FE/MD or proprietary-material (Digimat/Abaqus) set-ups; no open CAD; fields partly non-finite.
FreeCAD Code_Aster laminate + CalculiX comparison; Hannover milling	held	3	conflicting or unmatched source references - Laminate result units and touched solver objects unresolved; milling has measurements, not FEM.
Test-bench frame (FreeCAD/CalculiX)	held	1	no deterministic, mesh-independent numerical reference - Displacement is prescribed; the only response target is a coarse-mesh peak stress.
ACP composite kiteboard (Ansys tutorial)	held	1	needs a proprietary solver/material model to define the problem - Layup lives in ACP; 'SOLVED' archive lacks results; reference from a different software version.
PrePoMax tutorial geometries (12)	held	12	no deterministic, mesh-independent numerical reference - Idealised tutorial parts with plot-only results in a manual.
Modular helical marine anchor (OpenEI, 3 load sets)	held	3	CAD only in a proprietary native format - CAD only as DesignModeler/SolidWorks/Parasolid; analysis uses idealised equivalent members.
PyDPF composites examples (5)	held	5	CAD only in a proprietary native format - Native SpaceClaim geometry, ACP layups; parser failures on two projects.
Layered composite pipe (2 revisions)	held	1	conflicting or unmatched source references - Two conflicting saved revisions; orthotropic shell layup; IGES is a surface.
Pneumatic multi-link gripper (nonlinear contact)	held	3	supports/loads/materials not recoverable as CAD-face set-up - Joint/contact definitions not recoverable without the 2026R1 archive decoder; baseline/A1 have tables only.
Fin Ray 3-finger hyperelastic contact	held	1	no B-rep CAD (mesh, scan or paper geometry only) - Full three-finger CAD not published (one finger STEP + assembled mesh).
Bolted nozzle-flange sector (3 repos)	held	4	CAD only in a proprietary native format - DesignModeler-only CAD; README vs deck conflicts (units, materials, temperatures).
Formula SAE chassis (4 static cases)	held	4	CAD only in a proprietary native format - SpaceClaim-only beam CAD; decoded fields conflict with the published displacements.
Go-kart chassis modal/harmonic	held	1	CAD only in a proprietary native format - SolidWorks-only CAD; plot/report references.
IIT Mandi shear frame (4 modal variants)	held	4	CAD only in a proprietary native format - Abaqus CAE-only CAD; joint modelling and damage-case revision mapping unresolved.
Double-wishbone suspension	held	1	conflicting or unmatched source references - Supplied CAD revision cannot be matched to the baseline or redesigned stress state.
Perforated brake disc steady thermal	held	1	supports/loads/materials not recoverable as CAD-face set-up - Peak is a prescribed boundary value; convection/temperature face scopes must be mapped from the deck.
Horizontal pressure vessel (Nimonic 75)	held	1	supports/loads/materials not recoverable as CAD-face set-up - IGES surfaces only; saddle support scope undocumented.

collection	status	tasks / candidates	blocker / decision
Pneumatic cylinder cover	held	1	supports/loads/materials not recoverable as CAD-face set-up - Source support is a single fixed node plus frictionless node sets; not expressible as CAD-face BCs.
IC engine connecting rod (2 saved cases)	held	2	supports/loads/materials not recoverable as CAD-face set-up - Traction/support scopes must be mapped from the ANSYS deck node components to STEP faces.
Connecting rod (SolidWorks/ANSYS report)	held	1	conflicting or unmatched source references - STEP in inches with a 4.98 m extent; plots in CGS; support scope undocumented.
Twin-wheel nose landing gear	held	1	supports/loads/materials not recoverable as CAD-face set-up - Only a minimum safety factor is published; load/support scopes only in report figures; materials partly unconfirmed.
Single-cylinder crank assembly	held	1	supports/loads/materials not recoverable as CAD-face set-up - Material cards undocumented; thermal references conflict.
Ventilated brake rotor (thermal + structural)	held	1	supports/loads/materials not recoverable as CAD-face set-up - IGES faces with no solid; ten fixed faces and pressure scopes not identified.
Pipe-flange assembly (2 connection models)	held	2	CAD only in a proprietary native format - SolidWorks-only CAD; load scope undocumented; drawing inch/mm conflict.
Bolted shelf with pretension and contact	held	1	conflicting or unmatched source references - SpaceClaim-only CAD; saved results disagree with the report (0.028 vs 0.025 mm).
CubeSat frame archive	held	1	conflicting or unmatched source references - Archive geometry does not match the published CubeSat.
Elmer conjugate cooling (radiator, tube)	held	2	conflicting or unmatched source references - Set-up lives in native FreeCAD/Elmer objects; README vs native heat inputs differ; fields undershoot ambient.
Titanium wing static + prestressed modal	held	2	CAD only in a proprietary native format - DesignModeler-only CAD.
Wheel rim static + modal (5 analyses)	held	5	CAD only in a proprietary native format - SolidWorks/SpaceClaim-only CAD.
Quadcopter frame static/modal/harmonic	held	3	CAD only in a proprietary native format - Native assemblies only; harmonic load cards conflict with the report; modal uses point masses.
Curved steam pipe and elbow thermal	held	4	CAD only in a proprietary native format - SpaceClaim-only CAD; problem figures not complete drawings.
Seven-fin heat sink steady thermal	held	1	supports/loads/materials not recoverable as CAD-face set-up - Convection scope (41 of 55 faces) not identifiable; the published temperatures imply ~0.047 m^2 of convective area, less than the fin sides alone (0.054 m^2).
NX fin-array heat sink	held	1	CAD only in a proprietary native format - NX-only CAD; heat input and film coefficients absent.
Spur gear pair	held	1	supports/loads/materials not recoverable as CAD-face set-up - Load, material and contact law not specified.
Spring shock absorber	held	1	supports/loads/materials not recoverable as CAD-face set-up - Loads and supports not documented.
Screw jack	held	1	conflicting or unmatched source references - README and plot references conflict; native CAD only.
Knuckle joint	held	1	CAD only in a proprietary native format - Inventor-only CAD; STEP not uploaded.
Multi-leaf spring modal	held	1	conflicting or unmatched source references - Two conflicting frequency sets; surface-only IGES.
Steel dogbone static/buckling control	held	2	CAD only in a proprietary native format - Discovery-only CAD; supports undocumented.
Piston-pin-connecting rod	held	1	conflicting or unmatched source references - Part files missing from the assembly; README conflicts.
Piston thermal-barrier coating	held	3	conflicting or unmatched source references - Export/report revision mapping unresolved; obsolete result state.
Heat-assisted incremental forming	held	6	conflicting or unmatched source references - Partial CAM CAD only; conflicting job tables.
Kresling origami	held	1	CAD only in a proprietary native format - Inventor/Parasolid-only CAD; curve-to-case mapping unresolved.
Bolted plywood truss	held	1	supports/loads/materials not recoverable as CAD-face set-up - Idealised 2-D truss export with unknown units.
Haunched steel portal frame	held	4	CAD only in a proprietary native format - Native CAD only; inch/mm drawing conflict; load scope ambiguous.
Aquantis tidal turbine	held	1	conflicting or unmatched source references - CAD/FEM pairing not established (45 m CAD vs 41.5 m FEM).

collection	status	tasks / candidates	blocker / decision
Splined differential shaft	held	1	conflicting or unmatched source references - Report displacement exceeds the part length ~963x; conflicting reports.
Carbon-fibre UAV frame (62 bodies)	held	1	supports/loads/materials not recoverable as CAD-face set-up - Orthotropic properties and 132 bonded contacts not frozen; report-only references.
External gear-pump housing	held	1	CAD only in a proprietary native format - SolidWorks-only CAD; configuration/plot pairing unknown.
Ribbed mounting flange (drawing + 55-row table)	held	1	conflicting or unmatched source references - Drawing dimension conflicts; screenshot values match no table row; supports undocumented.
fcVM workbench examples: plastic peaks that are not limit loads, and post-buckling peaks (5 results)	held	5	no deterministic, mesh-independent numerical reference - The lug, barrier and train-bridge outputs end before the load-displacement curve flattens, so their peak is a lower bound, not a limit load; the plate-girder and U100 geometric-nonlinear peaks depend on fcVM's imperfection settings. The three plateau cases are studied under registry fcvm-plastic.
Paper drawings with FEM (pump shaft, ...)	held	10	no B-rep CAD (mesh, scan or paper geometry only) - Paper-only geometry and results; solver set-ups incomplete or inconsistent.
TU Wien e-bike frame	held	1	CAD only in a proprietary native format - SolidWorks-only CAD; plot-only, incomplete solver specification.

Files: candidate/ (give to the model), evaluator/ (keep hidden: references, registry, scorer, review protocol, evidence bundle), held/ (held tasks with their inputs and the excluded local targets), report/, scripts/. Licences differ per source and several sources are local-research-only: do not publish the package as a whole. No model results yet.